

Figure 2 — ML-KEM-1024 (Kyber-1024) Algorithm Flow [FIPS 203]

(a) KeyGen

Sample seed ρ



$A \leftarrow \text{XOF}(\rho) \in \mathbb{R}_q^{\{k \times k\}}$



$(s, e) \leftarrow B_{\{\eta_1\}}$



$t = A \cdot s + e \bmod q$



$pk = (A, t)$



$sk = s$

(b) Encaps

$m \leftarrow \{0,1\}^{\{256\}}$ (uniform)



$(K, r) = G(m \parallel H(pk))$



$(r_1, r_2, e_1) \leftarrow B_{\{\eta_1\}}$



$e_2 \leftarrow B_{\{\eta_2\}}$



$u = A^T \cdot r_1 + r_2 \bmod q$



$v = t^T \cdot r_1 + e_2 + \lfloor q/2 \rfloor \cdot m \bmod q$



$K_{enc} = \text{Compress}(u, v)$



$K = \text{KDF}(K)$

(c) Decaps

$\text{Decompress}(K_{enc}) \rightarrow (u, v)$



$m' = \text{Decompress}(v - s^T \cdot u \bmod q)$



Re-encapsulate with m'



Verify $K_{enc} == K_{enc}'$



If match: $K = \text{KDF}(K')$



If mismatch: $K = \text{KDF}(z)$
(implicit rejection)